Ou Ocean Kun Hei

HKUST(GZ) | Student ID: 50011920
BSc in Data Science and Big Data Technology
(Expected May 2027)

+86 15112134690
okhou533@connect.hkust-gz.edu.cn
https://github.com/oukunhei

Education

Hong Kong University of Science and Technology (Guangzhou) — 2023 - Present
Relevant Coursework: Deep Learning, Introduction to Natural Language Processing and Knowledge Engineering, Database Management Systems, Machine Learning, Discrete Mathmatics, Applied Statistics — CGPA: 3.362/4.4

Project Experience

LinguaSQL: Unlocking NL2SQL Potential Through Multilingual and Multi-Dialect Prompting — X Program, 2025.4 - 2025.12

- Developed a framework for generating candidate SQL queries with prompts in different natural languages and database dialects.
- Explored a selection strategy based on AST and gained approximately a 4% improvement in accuracy.

Large Model Self-Alignment Optimization System — Course Project, 2025.12

Reproduced the core process of the self-alignment paper, constructed a reverse model of "answer → instruction" through LoRA fine-tuning to generate synthetic data. Forming a complete optimization closed loop from data synthesis and quality evaluation to model iteration.

Data Science Project — Course Project, 2025.9 - 2025.12

Developed a personalized study-report feature in AnkiDroid open-source project, deploying a machine learning model via Flask. Implemented agile development and automated deployment using GitHub Workflow.

Campus Activities Registration and Management System — Course Project, 2025.4 - 2025.5

Responsible for core backend development, independently completed the design and implementation of the entire data access layer and business logic layer, ensuring the efficiency and stability of key operations such as activity lifecycle management.

Exploring Power Outages in the Context of Climate Change and Energy Transition — Summer Project, 2024.6 - 2024.9

Developed a climate-energy resilience model by analyzing global renewable energy and climate datasets in R, utilizing regression analysis to quantify future power outage risks under climate change scenarios.

Skills and Interests

Programming: Python, SQL, c++
Frameworks: PyTorch
Languages: English, Mandarin, Cantonese
Areas of Interest: Data Engineering, Retrieval Augmented Generation, NL2SQL

Volunteer Activities

PLACE Summer Camp, HKUST(GZ) — 2024, 2025
15th National Games of the People's Republic of China — 2025.11

Honors & Awards

Second-class Scholarship for Hong Kong, Macao, and Overseas Chinese Students — 2025
Mirit Research Award of Undergraduate Research Program — 2024
HKUST(GZ) Fisrt Class Scholarship — 2023